



Sneh Vijayvargiya

B.Tech

Indian Institute of Technology, Kharagpur

+91-7737748669

snehvjay@gmail.com

snehvijay2406@kgpian.iitkgp.ac.in

GitHub | LinkedIn

EDUCATION

Degree/Exam	Institute	Year
B.Tech	IIT Kharagpur	2026
RBSE XII	Arihant Public School, Kota	2021
CBSE X	Disha Delphi Global Public School, Kota	2019

PROJECTS

Custom GPT Architecture & Weight-Mapping

May'26

PyTorch-based 124M parameter generative transformer built from scratch

GitHub

- **Reduced trainable parameter to ~39.3M unique & ~78M total** by implementing weight-tying across the $50,257 \times 768$ tensor
- Build a tensor-mapping framework to match and inject pre-trained OpenAI GPT-2 weights into a custom 12-layer pipeline
- **Scaled autoregressive context windows upto 400% (256 to 1024 tokens)** by dynamically refactoring model dimensions

Mistral-7B Fine-Tuning Pipeline

May'26

An optimized LLM training framework leveraging parameter-efficient techniques for consumer hardware

GitHub

- Built a **4-bit QLoRA pipeline** for Mistral-7B via Unsloth, capping peak memory to **5GB VRAM** on a T4 GPU.
- Optimized an **SFT** using 8-step gradient accumulation to train **41.9M parameters (0.58%)** with stable convergence.
- Designed a **data pipeline** using sequence packing and multi-process tokenization to eliminate VRAM padding.

Interview Enhancer

Nov'25

A multi-modal AI system integrating advanced CV, real-time facial, and nuanced voice analysis

GitHub

- Made **4 ML pipelines** for ATS CV ranking, facial emotion analysis with CNN, **voice profiling via LSTM** & automated scoring
- Devised PyAudio pipeline with **tokenization, vectorization & LSTM** attaining **87% accuracy & 15%** higher test score
- Created a **contrastive learning, CNN** with **92%** accuracy for facial emotion recognition & Canny edge detection

Automated Insurance Data Entry System

May'25 – July'25

Business Solution Designer

GitHub

- Orchestrated an **agentic multi-agent** AI system using **LangChain** and **Llama 3.2** via **Ollama** for local, insurance data entry
- Developed an **Agent Orchestration** framework using **CrewAI** to manage user interaction and document processing workflows
- Implemented DocProcessor Agent utilizing **PyPDF2**, **Bedrock Embeddings**, and **ChromaDB** to vectorize and store documents
- Built an web application with a real-time chat interface using **Flask**, **PostgreSQL** (SQLAlchemy), and **HTML/CSS/JS**

TECHNICAL SKILLS

- **Languages & Tools:** Python, SQL, JavaScript, ChromaDB, Hugging Face, GitHub, CI/CD, FastAPI, QLoRA
- **AI & ML:** LangChain, CrewAI, PyTorch, RAG, TensorFlow, Keras, OpenCV, Unsloth, Supervised Fine-Tuning
- **Soft Skills:** Analytical Thinking, Leadership, Communication, Planning & Execution, Creative Problem Solving

KEY COURSES TAKEN

- **Engineering:** Programming and Data Structures | Probability and Statistics | Computational Hydrodynamics
- **MOOCs:** Data Structures and Algorithms | MERN Stack Development | Machine Learning(AZ) | Neural Networks and Deep Learning | Improving Neural Networks | Generative AI & LLM Specializations | MLOps & Systems

POSITIONS OF RESPONSIBILITY

- **Governor-Technology Comedy Club**, Technology Students' Gymkhaana May 2024 - April 2025
 - Conducted **4 different Flagship events** with a large expected footfall of around **3200** in total and **5+ successful collaborations**
 - Successfully increased the **Annual Budget** by around **33%**, **Content Engagement** by around **42%** and total posts by 40%
 - Collaborated with esteemed comedians **Piyush Sharma** and **Nishant Suri**, achieving a collective footfall of **5000+** audience

ACHIEVEMENTS

- **JEE Advance 2022:** Secured an All India Rank of 10593 in JEE Advanced 2022 out of 1,80,000+ students
- **JEE Mains 2022:** Secured a percentile of 99.21% in JEE Mains 2022 out of 12,00,000+ students

EXTRA CURRICULAR ACTIVITIES

- **MS Hall Tennis Team:** Served as Vice-Captain, for performance in achieving overall 4th position among 23 halls.
- **Mentor:** Got selected as a mentor for 5 fresher students and guided them to acclimate in the campus environment
- **MS Hall Wing Representative:** Represented the whole wing consisting of 24 Room Boarders in front of Hall Senate
- **Judge:** Judged PG Student's Event for different types of art forms with a total participation of over 50 students